

Introduction to TestSuite 7.0

WinRunner - Student Workbook



MERCURY INTERACTIVE

B *Appendix: Answers to Questions*

1—Recording and Playback



Why should you create a real test framework on a shared, network drive?

So that all members of the testing team have access to it and test collateral is backed up.

Exercise B: Set Initial Conditions



Describe the operation performed by each of the following statements:

```
set_window ("Flight Reservation", 10);
```

Sets the focus of the Flight Reservation window.

```
edit_set ("Date of Flight:", "12/12/03");
```

Types “12/12/03” in the date field.

```
list_select_item ("Fly From:", "Los Angeles");
```

Selects “Los Angeles” from the *Fly From* list.

```
button_press ("Flights...");
```

Presses the “Flights...” button

```
button_set ("First", ON);
```

Sets the state of a radio button or check button to ON.



Did the test play back to completion? If not, why not?

No. The *Insert Order* button was not enabled when the script tried to execute the line to push that button.



Were the initial conditions of the application the same when you recorded the test and when you ran it? How can you correct the problem?

No. When you initially recorded the test, no order was open in the window. However, when you ran the test an order was already open; therefore, the *Insert Order* button was disabled.

Correct it by ensuring that the test always starts by setting the initial conditions of the application. In this case you can start the test with a File > New selection.



Why is it a good idea to run the test twice?

You want to make sure that the test cleans up after itself, i.e., that it will run without error more than once.

Exercise C: Context Sensitive and Analog Recording Modes



Was the signature reproduced during playback?

No.



Was the signature reproduced during playback?

Yes. The signature that was reproduced during playback was the one recorded in Analog mode. The reason for this is that while Analog mode captures and replays the actual mouse movement (`move_locator_track`), Context Sensitive mode captures only the beginning (mouse downstroke) and end (mouse upstroke) positions of the mouse track.



Can you tell the difference between the Context Sensitive statements and the Analog statements? Which are more intuitive?

The Analog script is based purely on mouse movements and keyboard activities. A test created in Analog mode usually consists of three statements: `move_locator_track`, `mtype`, and `type`.

The Context Sensitive script is based on GUI objects and the actions performed on those objects. Context Sensitive statements usually start with the class of object that is being acted upon (e.g. `button_`), followed by the action being performed on that object (e.g. `button_press`).

Because of this, Context Sensitive is the more intuitive and readable of the two.



In what testing situation would it be appropriate to use analog recording mode rather than context sensitive?

For operations where the mouse tracks are important.



Why is it good practice to use the softkey to switch recording modes rather than the menu?

Because the mouse movements using the menu to switch modes get recorded in analog mode. Those movements are not part of the test.



Before switching to analog mode and signing your name, you were asked to move the window to a new position. Why is it important to move a window to a specific location before recording in Analog mode?

Because analog mode relies on absolute screen coordinates for accuracy. So, before you record in analog mode, you want to ensure that the window you perform operations on will be in the same position on the screen each time you run the test.



Was the signature reproduced during playback?

Yes.



Was the script easier to read than the one recorded in analog mode?

Yes.

Exercise D: The GUI Map - An Introduction



Did the script play back? If not, why not?

No. "WinRunner cannot find the window 'Flight Reservation.' "



Did the script play back?

Yes.



What is the general purpose of the GUI Map?

To translate object identifications between the AUT and the script.

Review: Record and Playback



1. **When would you record a test in Analog mode instead of Context Sensitive mode?**

When you need to capture the actual mouse movement. For example, when drawing a curved line in the “Paintbrush” application.



2. **Assume you are recording in Analog mode and you want to stop recording. Which method would you use: softkey, menu command or toolbar? Why?**

You would use the STOP softkey (Left Ctrl + F3). If you used the menu command or the toolbar, WinRunner would capture the mouse track used to select the command and thereby create unnecessary statements in your test script.

2—Synchronization

Exercise A: No synchronization



Did a WinRunner message appear? If yes, what is the message?

Yes. The message was: “Window: ‘Flight Reservation’; Object: ‘Delete Order’;
Error: Object is currently disabled.”

Exercise B: Synchronize Using obj_wait_bitmap



Did the test play back successfully this time?

Yes.



Why is an `obj_wait_bitmap` statement potentially unreliable?

Because it is dependent on screen resolution, font, color, etc.



Can you think of a more reliable method of synchronization?

Use a change in one of the attributes of the object.

Exercise C: Synchronize Using obj_wait_info



Why did you select the label attribute from the attribute list?

Because the label contained the text the test was supposed to wait for.



Will WinRunner wait longer than 20 seconds if it needs to?

Yes. It will wait 20 seconds plus the global timeout value.



What was the result of the modified test?

OK.

Review: Add Synchronization Points



1. In which cases do you need to insert synchronization points in your script?

You need to insert synchronization points into the script when WinRunner should wait for the application to complete an action, such as a database retrieval or query.



2. Which method would you use to synchronize the following:

a) Window: `set_window`

b) Bitmap image: Wait Bitmap command

c) Status bar message: `obj_wait_info` is recommended; or Wait Bitmap



3. What are the differences between `obj_wait_bitmap` and `obj_wait_info`?

`obj_wait_bitmap` waits for the a bitmap of a specified object, while `obj_wait_info` waits for an object, such as a label, to attain a specified value. The latter is more reliable because it doesn't rely on display settings.

Another difference is that `obj_wait_bitmap` causes a line to be written in the Test Results from which the captured image can be accessed. `obj_wait_info` does not appear in the Test Results.

4—Bitmap Checkpoints

Exercise: Bitmap Checkpoints



What does the image show?

It shows a blank rectangle (the *Agent Signature* area).



What was the reason for failure?

The *Clear Signature* button in Flight1B doesn't work. The *Agent Signature* area was supposed to be blank but it wasn't.

Review: Bitmap Checkpoints



- 1. Assume that you created a test with a bitmap checkpoint on one machine, and you run the test on another machine with a different resolution or color setup. What problem would you encounter? How can you solve it?**

Because of the change in resolution and colors, WinRunner will detect mismatches when there is actually no “bug” or real difference between the images. One way to correct the problem is to run the tests in Update mode on the second machine and recapture the expected results. In general, it is recommended that the testing team agree on a single video configuration for all machines.

5—GUI Checkpoints

Exercise A: Check Multiple Objects using a Checklist



What is the default check for each type of object?

Edit field - Compare as text

List field - Selection

Push button - Enabled



How can you test the checkpoint for both pass and fail conditions?

For the pass condition, execute the check just after selecting File > New. That will ensure the edit and list fields have no data in them and the push buttons will be disabled. For the fail condition, enter data in all fields, then run the check.



What is the difference between the first and second `win_check_gui` statements in the script?

The checklist `list1.ckl` uses the expected results file `gui1` in the first check and the expected results file `gui2` in the second check.



Did both checkpoints pass?

Yes.



Did the test pass? If not, why not?

The “Enabled” check on the *Flights* button failed. The button was expected to be disabled, but it was enabled instead.

Review: GUI Checkpoints



- 1. In this exercise you did not check any menu items. What would you check in a menu item?**

A menu item could be checked for its existence, its position within the menu, whether it is enabled or disabled, whether it has a submenu, and submenu count.

6—Verifying Text

Exercise A: Reading Text from a Bitmap Area



What statement was generated by WinRunner?

`win_get_text`



Was a comment generated? What were the contents of the comment?

Yes. “Version 1.0”.



Did Flight1B pass the test? If not, why?

No. A mismatch was discovered because the version number was missing from the About window in Flight1B.

Exercise B: Learn a Font



Did the comment “He is reading a book” appear in the script?

Yes.



Did WinRunner read the text?

Yes.

Review: Verifying Text



- 1. When would you read text using Get Text commands rather than using Context Sensitive functions?**

You would use Get Text Commands in cases where you want to read text from a non-GUI-object area of your application, such as bitmap or ASCII text.

8—Managing GUI Map Files

Exercise A: Learn the Windows/Objects of the AUT



List the GUI Map files currently open or loaded.

`flight70.gui` and `<Temporary>`



To which GUI Map File were all the learned objects added?

Temporary.



What is the name of the *Flights* button in the GUI Map File?

FLIGHT



List the attributes of the *Flights* button that WinRunner learned (look at the *Physical Description*).

class: `push_button`

label: FLIGHT

Exercise B: Modifying GUI Files



Which Flight Reservation button is `Button_3`? What would be a more descriptive name for this button?

Open Order button on the toolbar. *Open Order - toolbar*.



What is the benefit of giving descriptive names to objects in the AUT?

It makes scripts more readable. It makes objects easier to find.



What objects in the Flight Reservation window are associated with the following logical names in the temporary GUI file?

OBJECT	MODIFIED NAME
Button	Reports - toolbar
Button_1	Graphs - toolbar
Button_2	Delete Order - toolbar
Button_3	Open Order - toolbar
Button_4	New Order - toolbar
Button_5	Help - toolbar



What visual cue indicates that a GUI Map File has been changed but not saved?

An asterisk [*] precedes the file name in the GUI File list when the file contains an unsaved change.



Did the modified name of Button_3 appear in the script?

Yes.



What new windows/objects were learned during recording?

Open Order window, *Edit* edit field, *OK* button, and *Order No.* check box.



Did the Open Order window and all its objects move to the lab11.gui file?

Yes.

Exercise C: Loading and Unloading GUI Map Files



What visual cue is displayed on the GUI Map Editor when a GUI Map file is loaded? What is the visual cue when a GUI Map file is open but not loaded?

When loaded, an “L” appears to the left of the GUI Map File name in the Editor. When open and not loaded, the file name appears in the Editor without the “L.”



What TSL statement can you use to unload and close all GUI Map files which may be open? Where should you place this statement in the script?

`GUI_close_all();`

At the top.



What TSL statement can you use to load a GUI Map file automatically from your script? Where should you place this statement in the script?

`GUI_load("<path + file name>");`

It should be the first statement after the `GUI_close_all` statement.



How do you know that the GUI Map File was loaded properly?

WinRunner was able to play back the script without error.

Exercise D: Fixing a GUI Map File



Did the script replay without error? If not, what was the error?

No. "WinRunner cannot find the push_button 'FLIGHT'".



In the boxes below, write the description of the *Flights* button as shown by the GUI Spy and by the GUI Map Editor.

DESCRIPTION - IN GUI SPY	DESCRIPTION - IN GUI MAP EDITOR
<pre>{ class: push_button label: FLIGHT }</pre>	<pre>{ class: push_button label: Flights }</pre>



Explain how you would fix the GUI Map file.

Use the GUI Spy to retrieve the correct description of the Flights button from the application.

Use the GUI Map Editor to change the description to match the application and to save the edited file.



Did the test play back successfully?

Yes

Exercise E: Adding the GUI Map Editor to the Custom User Toolbar



In what category would the GUI Map Editor be found?

Tools



Which command refers to the GUI Map Editor?

Edit GUI Map



Was a GUI Map Editor icon added to your toolbar?

Yes

Review: Managing the GUI Map



- 1. Assume you are about to begin creating automated tests for an application. Would you consider using one GUI Map file per test instead of one GUI Map file for the entire application? Please explain your answer.**

Yes, because it is easier to maintain.



- 1. When do you need to update the GUI map file?**

You need to update the GUI Map when objects in the application have changed. This usually happens when a new, major version of the application is released.



2. Select the correct statement.

When a new version of the application is released and has changes in the user interface:

(c) The GUI map file from the previous version can be modified according to the changes in the application.

9—Batch Testing

Exercise B: Create the Main Test



What statement should be used to call the login test?

```
call login();
```



How should we modify this `call` statement to pass your name and the password "mercury" to the login test as parameters?

```
call login("joe", "mercury");
```



What statement will call the open_order test?

```
call open_order();
```

Exercise C: Run Batch Test



WinRunner cannot find a checkpoint's expected results. Why not?

Because the checklists and expected results are saved with the called tests, not in the directory of the main test.



There are two solutions to this problem. What are they?

Turn on batch mode or manually copy the checklists and expected results files to the main test's directory.

10—Practice and Test Case

Practice Case



Which method of verification will you use to verify the initial state of the objects in the Search Results window?

With a `win_check_gui()` inserted with the GUI Checkpoint > For Multiple Objects menu selection.



Which method of verification will you use to verify that the Fax Preview window displays the correct order?

With a `win_get_text()` inserted with the Get Text > For Screen Area menu selection.



How will you ensure that the agent signature is put in exactly the same location each time the script is run?

When recording, move the window just before writing the signature.



Which method of verification will you use to verify that the Fax Preview window displays the agent signature?

With a `obj_check_bitmap` inserted with the Bitmap Checkpoint > For Object / Window menu selection.



What visual cue are you waiting for after pressing the OK button on the Login window?

The appearance of the Flight Reservation window.



How will you synchronize the test with the AUT after logging in?

With the statement: `set_window("Flight Reservation", 40);`



How will you synchronize the test with the AUT after sending the fax?

With an obj_wait_info on the progress bar message: "Fax Sent Successfully..." inserted with the Synchronization > For Object / Window Property menu selection.



What defect(s) did you find in Flight1B as a result of this test?

The *Send Signature with order* button in the Fax Order window did not work. The bitmap checkpoint failed because the signature did not appear in the Fax Preview window.



Which attribute did you use to check that an order for customer Kim Smith appeared in the Search Results window?

Content.

Appendix A—RapidTest Script Wizard

Exercise: Create the GUI Map using the RapidTest Script Wizard



What is the actual object in the application? Suggest a more descriptive name for this object.

“New Order”. “ToolBarNew”



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Training Evaluation Form

COURSE: WinRunner C/S Basic Training (7.0/01) NAME: _____

DATE: _____

COMPANY: _____

INSTRUCTOR: _____

YOUR DEPARTMENT: _____

TELEPHONE #: _____

1. How would you rate the quality of the instructor's presentation? Please circle only one rating for each category:

	Excellent		Good		Average		Poor	
Stated objectives clearly	8	7	6	5	4	3	2	1
Displayed knowledge of subject	8	7	6	5	4	3	2	1
Encouraged expression of opinions/ideas	8	7	6	5	4	3	2	1
Understood and responded to questions	8	7	6	5	4	3	2	1
Showed enthusiasm	8	7	6	5	4	3	2	1
Spoke understandably (clarity, volume, speed)	8	7	6	5	4	3	2	1

Additional comments/suggestions about the instructor's presentation skills.

2. How would you rate the quality of the course materials (slides and student workbook)? Please circle only one rating for each category.

	Excellent		Good		Average		Poor	
Overall effectiveness	8	7	6	5	4	3	2	1
Logical organization	8	7	6	5	4	3	2	1
Supported the course objectives	8	7	6	5	4	3	2	1
Comprehensive	8	7	6	5	4	3	2	1
Layout / visual clarity	8	7	6	5	4	3	2	1

Additional comments/suggestions about the course materials. Please list three specific suggestions.

1. _____
2. _____
3. _____

(PLEASE TURN OVER)

3. Did you find any of the exercises or review questions difficult? If so, specify which topics could have been explained more clearly.

4. Is there any topic that you feel should have been presented in more depth? If so, please specify.

5. What were the most important skills, concepts, or ideas you learned in these courses?

1.

2.

3.

6. Were your needs and objectives met? Why or why not?

7. Would you recommend our products to others? Why or why not?

8. Would you act as reference to other customers? Yes No (If not, please explain)

9. Are you interested in any other Mercury Interactive products? If so, which one(s)?

10. How would you rate the course administration? (for public classes only) Please circle only one rating for each category:

	Excellent		Good		Average		Poor		N/A
Registration procedures	8	7	6	5	4	3	2	1	—
Classroom	8	7	6	5	4	3	2	1	—
Course equipment (computers, projector, etc.)	8	7	6	5	4	3	2	1	—
Food	8	7	6	5	4	3	2	1	—

Additional comments/suggestions about the course administration:
